

Task 9 - Game Data Structures Short Report (Vectors)

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COS30031 Games Programming

In creating the Inventory system, I have opted to use Vectors data structure from the C++ Standard Library as after I evaluated other data structures, I concluded that we would need to take performance and usability into consideration for our inventory system.

Firstly, in games inventories are dynamic in nature and the number of items that can be stored in them varies. Therefore, using a structure such as vector can be designed to grow or shrink as needed which makes it a good choice for this scenario. Moreover, its allocated memory in blocks which lead to enhanced performance due to better cache coherency.

Therefore, It's efficiency should not be underestimated as players that add, remove or view inventory should be swift as we benefit from $O(1)$ access times for indexed elements and ensures quick access to items.

However, no data structure is perfect as one of the concerns of vectors is that the removal of items from the middle of a vector can have an $O(n)$ complexity that stems from the need to shift elements which can cause blocknecks if removal of items is frequent. In addition, vectors are designed to be dynamic and sometimes this can cause more memory to be allocated than needed which can cause memory overhead. In comparison, other data structures such as a list which has memory overhead where each element requires additional memory due to it storing two pointers that are the previous element and next element which can lead to increased memory overhead compared to vectors. Moreover, list does not store its element in contiguous memory which can result in poor cache locality compared to vector which stores in contiguous memory.

In conclusion, vector data structure, in context for an Inventory System can be an efficient solution and benefits for typical game scenarios such as in RPG games where massive inventory systems are used and the positives outweigh the negatives as frequent removal does not happen.